

The Period Relation of the One-Ring Read-Head

Why $(n, b) = (8, 16)$ Is the Only Integer Radix, and Why That Is Not Yet the Whole Reason

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Status of Every Claim

§	Statement	Status
1	Single damped pole pair, harmonic weights, integer radix, shell closure $\Rightarrow (\theta, r, b) = (\pi/4, 1/\sqrt{2}, 16)$	THEOREM — proved
2	The read-head extracts a rational multiple of $\pi \Leftrightarrow \pi \in \mathbb{Q}\text{-span}\{l_a\}$; admissible coefficients form the residue image $\mathcal{R}_{n,m}$, not $\mathbb{Z}^n$	THEOREM — reduction
3	Across the tested grid, the period relation holds only at $(8,16)$	MEASURED — bounded height
4	Angle closure, log closure, and torsion are each necessary and none sufficient	COUNTEREXAMPLES
5	The $\sqrt{2} / \zeta_8 / 15$ alignment explains $(8,16)$	MEASURED
6	The period relation holds only at $(8,16)$ for all n	CONJECTURE — open

Nothing in this paper is asserted at a status above the one in that table. Section 3's exclusions are bounded-height statements verified to coefficient bound 10^6 ; they are not proofs of non-existence and are never used as such.

1. Theorem 1 — Uniqueness of the Integer Radix

Assumptions, stated so the theorem's scope cannot drift. A1: the readout is harmonic weights $w_m = 1/m$ on a single damped pole pair. A2: the observable is π , giving the design curve $r(\sin\theta + \cos\theta) = 1$. A3: the shell closes, $\theta = 2\pi/n$ with n a positive integer. A4: the radix is an integer, $r^n = 1/b$ with $b > 1$. A5: the series converges, $r < 1$.

A2 with A5 gives $\sin 2\theta > 0$, hence with A3, $n \geq 5$. A2 gives $r^2 = 1/(1 + \sin 2\theta)$, so A4 requires

$$b(n) = (1 + \sin(4\pi/n))^{n/2} \in \mathbb{Z}, \quad b > 1$$

1.1 Integrality forces rationality

Let $s = \sin(4\pi/n)$. Every Galois conjugate of s is a value $\sin(4\pi k/n)$, hence real and in $[-1, 1]$, so $1 + s^\sigma \in [0, 2]$ for every σ . If n is even, $b \in \mathbb{Q}(s)$; if $b \in \mathbb{Z}$ then b is Galois-fixed and $(1 + s^\sigma)^{n/2} = (1 + s)^{n/2}$ for every σ . For odd n apply the same to $b^2 = (1+s)^n \in \mathbb{Z}$. Since $t \mapsto t^{n/2}$ is strictly increasing on $[0, \infty)$, two non-negative reals with equal positive powers are equal, so $1 + s^\sigma = 1 + s$ for every σ , hence $s \in \mathbb{Q}$.

This step is the one that was missing from the first version of this argument, where integrality was assumed to force rationality without justification. It does force it — but through strict monotonicity of $t \mapsto t^{n/2}$ on the non-negative reals, not directly.

Niven's theorem now applies: a rational sine at a rational multiple of π lies in $\{0, \pm 1/2, \pm 1\}$. Five cases remain.

s	$4\pi/n$	n	$b = (1+s)^{n/2}$	Verdict
1	$\pi/2 + 2k\pi$	8	16	ADMISSIBLE
1	$\pi/2 + 2k\pi$	8/5, 8/9, 8/13	—	n not an integer
1/2	$\pi/6, 5\pi/6$	24	531441/4096	b not an integer
1/2	$\pi/6, 5\pi/6$	24/5, 24/13, 24/25	—	n not an integer
0	0 or π	4	1	$n < 5$, and $b = 1$
-1/2, -1	—	—	—	$\sin 2\theta < 0$, violates A5

Exactly one survivor: $\theta = \pi/4$, $n = 8$, $r = 1/\sqrt{2}$, $b = 16$. The failure elsewhere is not marginal — at $n = 12$, $s = \sqrt{3}/2$ and its conjugate $-\sqrt{3}/2$ give $(1+s)^6 = 42.218744$ against $(1-s)^6 = 5.7827543e-6$, so $b(12)$ is not even rational.

$$b(8) = (1 + \sin(\pi/2))^4 = 16.0$$

$$h'(x) = x/(\sin(x) + 1) \quad [\text{monotonicity identity, symbolic}]$$

conjugates of $\sin(4\pi/n)$, $n = 5..30$: ALL real and in $[-1, 1]$

Niven survivors: [(8, '16', integer=True, sin>0=True)]

The notebook verifies the total-reality of the conjugate set computationally for $5 \leq n \leq 30$. That range is verification of the symbolic step, not the proof; the proof carries the general Galois argument above.

A remark on what fell away. Earlier versions of this argument leaned on a monotonicity lemma for $b(n)$, an explicit bound $b(n) < e^{(2\pi)}$, the transcendence of $e^{(2\pi)}$ by Gelfond–Schneider, and a finite exhaustion over 42,970 shell counts. All four are correct and none is needed. The Galois step removes the search entirely.

2. Theorem 2 — Reduction to a Period Relation

Write the read-head content as an integral basis. For $D(t) = b - t^n$ set $I_a = \int_0^1 t^a / (b - t^n) dt$ for $a = 0 \dots n-1$. A numerator $N \in \mathbb{Z}[t]$ of degree $< n$ gives

$$\int_0^1 N(t)/(b - t^n) dt = \sum_a n_a I_a = \sum_j C_j \log \alpha_j, \quad C_j = -\rho_j N(\rho_j)/(nb)$$

where ρ_j are the roots of $b - t^n$ and $\alpha_j = 1 - 1/\rho_j$. The read-head extracts a rational multiple of π if and only if

$$\pi \in \mathbb{Q}\text{-span} \{I_0, \dots, I_{n-1}\}$$

This is the spine of the paper. Two things follow immediately and both matter.

2.1 The coefficient space is not $\mathbb{Z}^n$

The residue coefficients C_j are not free. They are the image of the integer numerator lattice under a Vandermonde map:

$$\mathcal{R}_{n,m} = \{ (-\rho_j N(\rho_j)/(nb))_j : N \in \mathbb{Z}[t], \deg N < n \}$$

An instrument built on this denominator can only reach coefficient vectors inside $\mathcal{R}_{n,m}$. Symmetries present in the pole group but outside $\mathcal{R}_{n,m}$ are unreachable, and Section 4 gives a computed witness of exactly that.

2.2 The branch-stable object is the sum, not the product

It is tempting to exponentiate and ask whether $\prod \alpha_j^{C_j}$ is a root of unity. For BBP the content is π itself, so the product is $e^{\pi} \approx 23.14$ — in the free part, not torsion. The exponentiated form also loses branch information. The period sum $\sum C_j \log \alpha_j$ is the stable object and is what this paper uses throughout.

Dual verification at (8,16), integral basis against pole decomposition:

```
integer numerator vector w = [64,0,0,-32,-16,-16,0,0]
```

```
sum_a w_a I_a = 3.141592653589793238462643
```

```
matches pi : True
```

```
residue form sum_j C_j log(1 - 1/rho_j)
```

```
real part = 3.1415926535897932385
```

```
imaginary residual = -4.01e-80
```

```
reldim(l) = 1 [one zero relation]
```

```
reldim(l, pi) = 2 [pi enters]
```

π in span : True

The jump from 1 to 2 on adjoining π is the observable. $\text{reldim}(I) = 1$ is the zero-relation line; the increment is the π -image. This decomposition — void dimension = π -image + zero-relation space — was derived independently in the companion work on pole annihilation and returns the same numbers here from a different setup.

3. Proposition 3 — Measured Scope

Relation dimension with and without π , across a curated grid of (n, b) . Computed at 120 digits.

(n, b)	b	$\text{reldim}(I)$	$\text{reldim}(I, \pi)$	π in span
(8, 16)	2^4	1	2	YES
(8, 4)	2^2	0	0	no
(8, 8)	2^3	0	0	no
(8, 9)	3^2	0	0	no
(8, 32)	2^5	0	0	no
(8, 64)	2^6	0	0	no
(6, 4)	2^2	0	0	no
(6, 64)	2^6	1	1	no
(12, 9)	3^2	0	0	no
(12, 16)	2^4	0	0	no
(12, 64)	2^6	1	1	no
(4, 16)	2^4	0	0	no
(10, 16)	2^4	0	0	no
(6,48) (8,48) (12,48)	composite	0	0	no
(8,6) (6,12) (8,36)	composite	0	0	no

3.1 Robustness of the negative entries

The exclusions are negative PSLQ results and therefore only as strong as the coefficient bound. They were re-run at bounds 200, 10^4 and 10^6 :

```
(n,b) maxc=200 maxc=10^4 maxc=10^6
(8, 16) 1->2 pi=Y 1->2 pi=Y 1->2 pi=Y
(6, 64) 1->1 pi=N 1->1 pi=N 1->1 pi=N
(12, 64) 1->1 pi=N 1->1 pi=N 1->1 pi=N
(8, 9) 0->0 pi=N 0->0 pi=N 0->0 pi=N
(8, 4) 0->0 pi=N 0->0 pi=N 0->0 pi=N
(12, 16) 0->0 pi=N 0->0 pi=N 0->0 pi=N
(10, 16) 0->0 pi=N 0->0 pi=N 0->0 pi=N
```

Stable across a 5000-fold increase in coefficient height. The claim is therefore: no period relation exists at these (n,b) with coefficients bounded by 10^6 . It is not a claim that none exists.

4. The Shadow Ledger

Four increasingly restrictive conditions, each necessary, each computed, and each killed as sufficient by an explicit witness. This section is the evidence that the period relation sits strictly below all of them.

4.1 Angle closure is not sufficient

At $(8,9)$ the arguments of the α_j satisfy an integer relation with π :

```
PSLQ on [arg a_1, arg a_2, arg a_3, pi] at (8,9): [-6, 0, 6, 1]
```

The angular channel closes. The period relation does not hold: reldim goes $o \rightarrow o$. Angle closure is abundant and selects nothing.

4.2 Angle and magnitude closing separately is not sufficient

At $(6,64)$ both channels close, independently:

```
log channel PSLQ on [log|a_o..a_3|] : [-1, 2, 0, -1]
```

```
arg channel PSLQ on [arg a_1, arg a_2, pi] : [-6, 0, 1]
```

Both lattices exist. π is still not in the span. The two channels closing as separate lattices is not the same as the coupled period relation closing, and this is the witness that separates them.

4.3 Torsion is not sufficient — the decisive counterexample

The same $(6,64)$ carries genuine torsion in the pole group:

```
arg(alpha_1)/pi at (6,64) = 0.1666666667 = 1/6 exactly
```

```
hence alpha_1/alpha_5 = e^{i pi/3}, a sixth root of unity
```

```
PSLQ on [l_o..l_5, pi] at maxcoeff 20000:
```

`[16, -24, -8, -6, 1, 0, 0] carries_pi = False`

Torsion is present and there is still no read-head. This kills the most elegant available formulation — that the gate is membership of $\prod \alpha_j \wedge C_j$ in μ_∞ — and it kills it with a case that passes every weaker test.

The relation the search does return is worth noting: its final coefficient is zero, so it is a relation among the I_a alone. It is the zero-relation line, and its first six entries are exactly the zero-relation vector obtained independently for $64 - t^6$ in the companion work.

4.4 The mechanism — why the instrument cannot reach the torsion

Solve the Vandermonde system for the numerator that would realise the torsion coefficient vector at (6,64):

`max |Im(w)| = 55.42562584220407`

The required numerator is not real, let alone integral. The symmetry exists in the pole group; the residue image $\mathcal{R}_{n,m}$ cannot address it.

The pole group contains the symmetry. The numerator lattice determines whether the instrument can couple to it. That is the physical content of Theorem 2.1, and 55.43 is its receipt.

5. The Alignment at (8, 16)

Two products appear in this construction and they are different objects. Conflating them would break the boundary identity, so both are stated.

`prod_j (sqrt2 - zeta_8^{j-j}) = 15.0 [boundary norm]`

`prod_j alpha_j = 0.9375 = 15/16`

`because alpha_j = 1 - zeta_8^{j-j}/sqrt2 = (sqrt2 - zeta_8^{j-j}) / sqrt2`

`and (sqrt2)^8 = 16`

`identity 15 = (15/16) * 16 : True`

`alpha_1 = (1 + i)/2`

The 15 is the boundary norm and it is the same 15 as $A(1) = (1+2)(1+2+2) = 3 \cdot 5$ in the Boundary Evaluation Identity of the companion work, where A is the annihilator $\gcd(N, D)$ and $\sum n_a = N(1) = A(1) \cdot M(1)$. The same algebraic object appears in the annihilator, in the pole product, and in the residue decomposition. The S -units involved are $\{2, 3, 5\}$.

The shell radius $\sqrt{2}$ lands the construction inside $\mathbb{Q}(\zeta_8)$, where the eighth roots of unity and $\sqrt{2}$ generate the same field. That coincidence is the explanation of the alignment. It is not the definition of it — the definition is the period relation of Theorem 2, and Section 4 shows every attempt to promote the arithmetic anatomy into the gate over-counts.

6. Conjecture and the Open Beam

CONJECTURE. For all n and all integer $b > 1$, the period relation $\pi \in \mathbb{Q}\text{-span}\{I_a\}$ holds only at $(n,b) = (8,16)$. Equivalently, $\mathcal{R}_{n,m}$ meets the π -locus only there.

For each fixed (n,b) the question is finite by S -unit finiteness once phrased as a linear-forms-in-logarithms problem: the α_j are S -units for a finite S determined by b and n , and the period relation asks whether a specific $\mathbb{Q}$ -linear combination of their logarithms equals a rational multiple of π . Bounding such a form is Baker theory. The global exclusion over all (n,b) is the open beam and it is Baker-hard. It is not nearly done.

What would settle the conjecture, in order of increasing difficulty: an effective height bound making the per- (n,b) check rigorous rather than bounded-height; a structural argument showing $\mathcal{R}_{n,m}$ misses the π -locus whenever the shell radius $b^{1/n}$ is not $\sqrt{2}$; and a general linear-forms bound covering all n at once.

7. What This Paper Does Not Claim

It does not claim BBP is the only base-16 formula for π . It does not claim the resonator and BBP are the same object — the pole correspondence $z = 1/\text{conj}(t)$ is a theorem, and calling that identity is an interpretation. It does not claim any physical system implements this readout. It does not claim the exclusions in Section 3 are proofs of non-existence. Theorem 1 is uniqueness within the family A1–A5 and nowhere else.

The one sentence that carries the paper: the geometry supplies a smooth one-parameter family and distinguishes no point on it. Every constraint that selects is arithmetic.

Appendix — Reproduction

All figures are the notebook's own output. Numerical work: mpmath at 80–120 decimal digits. Symbolic work: SymPy exact arithmetic. Integer relation detection: PSLQ, with negative results re-verified at coefficient bounds 200, 10^4 and 10^6 . The companion computations on pole annihilation, the void dimension and the Boundary Evaluation Identity were run in a separate session on a different setup and agree numerically where they overlap.